

Erasmus Mundus Joint Master in Manufacturing 4.0 by intElligent and susTAinable technologies



MASTER's Degree Thesis

*Identification by inverse method and data analysis of
mechanical behaviour laws dedicated to the numerical
simulation under machining conditions.*

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Abstract:

This thesis presents an inverse identification method for calibrating material behavior laws to enable accurate numerical simulation of machining processes. It focuses on the orthogonal cutting of 15-5PH martensitic stainless steel, the study combines finite element modelling in Abaqus with a Python-driven optimization loop using the Levenberg-Marquardt algorithm. The approach systematically adjusts Johnson-Cook constitutive parameters and the Taylor-Quinney coefficient to minimize discrepancies between simulation outputs and experimental reference data, specifically chip thickness, tool-chip contact length, cutting force, and penetration force across five undeformed chip thicknesses.

Initial simulations using reference parameters showed significant inaccuracies, with average errors of 21.94% (chip thickness), 41.20% (contact length), 40.21% (cutting force), and 28.37% (penetration force). After optimization, these errors



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were drastically reduced to 0.49%, 4.35%, 0.34%, and 0.14%, respectively. The algorithm demonstrated rapid and stable convergence, typically within five iterations per case. The resulting parameter set not only improved predictive accuracy in 2D orthogonal cutting but also generalized effectively to 3D validation, confirming its readiness for industrial application in virtual machining and process optimization.

Keywords: (Inverse Method, Machining, Mechanical Behaviour Laws, Numerical Simulation)

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